

Assignment 1

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Question 1: Initial AddRoundKey

`round[ 1].start` 00102030405060708090a0b0c0d0e0f0

Question 2: SubBytes

`round[ 1].s_box` 63cab7040953d051cd60e0e7ba70e18c

Question 3: State Matrix

`round[ 1].state matrix (Q3):`
63 09 cd ba
ca 53 60 70
b7 d0 e0 e1
04 51 e7 8c

Question 4: ShiftRows

`round[ 1].s_row` 6353e08c0960e104cd70b751bacad0e7

Question 5: MixColumns

`round[ 1].m_col` 5f72641557f5bc92f7be3b291db9f91a

Question 6: AddRoundKey (Round 1)

`round[ 2].start` 89d810e8855ace682d1843d8cb128fe4

Question 7: Rounds 2–9

`round[10].start` bd6e7c3df2b5779e0b61216e8b10b689

Question 8: Final Round

output (ciphertext) 69c4e0d86a7b0430d8cdb78070b4c55a

Bonus: Decryption

bonus ioutput (plaintext) 00112233445566778899aabbccddeeff